Applied Information Extraction in Python

Today's goal • 2/3 items

Module 1
Information Extraction

Welcome to the Course

Welcome to Information Extraction
Video • 6 min

MADSwPy Certificate Roadmap
Reading • 10 min

Course Syllabus
Reading • 10 min

Introduction to Jupyter Notebook
Reading • 10 min

Meet Other Learners
Discussion Prompt • 10 min

Help Us Learn About You
Reading • 10 min

Introduction to Information Extraction

What is Information Extraction?
Video • 8 min

Information Extraction in Different Domains
Video • 6 min

Knowledge Check: Introduction to Information Extraction
Practice Assignment • 10 min

Rule-Based Approaches to Information Extraction

Extracting Formatted Information
Video • 7 min

Regular Expressions in Detail
Reading • 20 min

Lookup Based Extraction
Video • 4 min

Knowledge Check: Rule-Based Approaches to Information Extraction
Practice Assignment • 10 min

How to Set Up Rule-Based Approaches in Python

Demo: Using Regular Expressions & Examining Output
Video • 4 min

Jupyter Notebook Practice on Basic NLP and Rule-Based Extraction
Lab • 1h

Module Wrap-Up & Assignments

Assignment 1 Introduction: Formatting & Normalizing Data with Regular Expressions
Video • 4 min

Build an Information Extraction Pipeline for Template/List-Based Fields
Programming Assignment • 3h

Module 1 Assignment
Graded Assignment • 30 min

Module 2
Named Entity Recognition (NER)

Module 3
Neural Network Models

Module 4
Transformers Transform Information Extraction

Introduction to Jupyter Notebook

0:13 / 3:26 2x

This introduction provides you with the basic instructions on how to effectively navigate and use Jupyter Labs on Coursera, ensuring you can make the most of your learning experience in this powerful space.

What is Jupyter Labs?

In this course, you will execute code directly on the Coursera platform using Coursera Labs. The labs are set up in an environment that leverages Jupyter Notebooks to present **assignment instructions** and **read-only pre-written code for execution**. This environment includes all necessary packages to run the pre-written code and most types of code relevant to the course. It is a ready-to-use workspace for learners to engage with the course material.

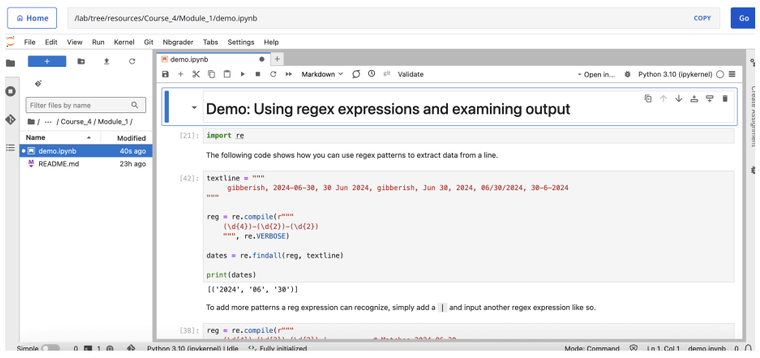
Note that as this platform is hosted in a shared cloud environment, some code may take longer to execute than it would if learners would execute the code themselves on their local machine. The Coursera Labs environment provides a convenient workspace for learners to write and execute code in, however the resources available to execute learner code on are fairly limited. Please be aware that some code segments, particularly those that utilize machine learning libraries, may take 30 minutes or longer to execute in full.

Navigating Jupyter Labs

Jupyter lab space includes two key components as shown in Figure 1:

- **A file browser** located on the left-hand side, and
- **A notebook interface** located on the right

Figure 1. A screenshot of the Jupyter lab environment



In this course, you will work in the notebook for the most of the time, especially in the tabs ending with ".ipynb". This file extension stands for "Interactive Python Notebook". Within this interface, you will notice code cells and, sometimes, markdown (text) cells. Here's how to work with them:

1. **Code Cells** contain Python code that you can execute to see the results immediately.
2. **Markdown (Text) Cells** contain text, instructions, or explanations. They are not meant for executing code but they provide important context and information.

To effectively use the programming exercises included in this course, first review the "implementation" document that we have provided before each programming exercise. This document explains different parts of the code and the output.

Run Pre-Written Code Cells

After reviewing the Implementation document, run the code in the Jupyter notebook. To run the code cells:

1. Select the Cell: Click on the cell to select it.
2. Execute the Cell: Use any one of the following methods to execute the selected cell.
3. Use one of these methods:
    - Press "Shift + Enter" keys to run the cell and move to the next one.
    - Press "Ctrl + Enter" to run the cell and stay in the same cell.
    - Click the "Run" button in the toolbar (represented by a "play" icon).

Running Code Cells

Some cells allow you to modify the code and experiment with your own implementations. Follow the assignment instructions, you will be asked to paste your code into these cells and run them to see the results.

1. Edit the Code: Click on the editable cell and modify the code as per the instructions.
2. Run the Cell: Use any one of the following methods to execute the selected cell.
    - Press "Shift + Enter" to execute the modified code and move to the next cell.
    - Press "Ctrl + Enter" to execute the code and stay in the same cell.
    - Click the "Run" button in the toolbar (represented by a "play" icon).

Go to next item ✓ Completed

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